

Zilong Li

GitHub: <https://lzlcfd.github.io/>

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SPECIALIZATION

Aerospace Engineering with a focus on
Computational Fluid Dynamics (CFD), Machine Learning and Reduced Order Modeling.

EDUCATION

Iowa State University (ISU) 08/2021-Present
Ph.D. in Aerospace Engineering Major GPA: 4.0/4.0; Overall GPA: 4.0/4.0

The Pennsylvania State University, University Park (PSU) 08/2017-12/2019
M.S. in Civil Engineering Major GPA: 3.96/4.0; Overall GPA: 3.92/4.0
M.S. minor in Computational Science GPA: 4.0/4.0

Shanghai Jiao Tong University (SJTU) 09/2012-03/2015
M.S. in Fluid Mechanics Major GPA: 87/100; Overall GPA: 85/100

Wuhan University of Science and Technology (WUST) 09/2008-06/2012
B.S. in Engineering Mechanics Major GPA: 89/100; Overall GPA: 87/100 Rank: 3/40

RESEARCH EXPERIENCE

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| 08/2021-Present | Department of Aerospace Engineering, ISU
Research Assistant in <i>Aerospace Engineering</i> <ul style="list-style-type: none">● Lead a project on Reduced Order Modeling (ROM)● Implemented and validated reduced order models in OpenFOAM● Machine Learning enabled turbulence models for unsteady aerodynamic flows |
| 08/2017-12/2019 | Department of Civil and Environmental Engineering, PSU
Research Assistant in <i>Civil Engineering</i> <ul style="list-style-type: none">● Collaborated on a project on sediment transport● Implementation of Smoothed Particle Hydrodynamics (SPH) method in DualSPHysics for granular flow● High-fidelity numerical simulation and analysis of turbulence |
| 06/2015-12/2015 | Estuarine and Coastal Science Research Center, Shanghai
Research Assistant in <i>Computational Fluid Dynamics</i> <ul style="list-style-type: none">● Lead a field investigation of flow around groins in Hangzhou Bay● Completed experimental investigation of flow around artificial groins and numerical simulation of the flow field● Experience with High Performance Computing (HPC) |
| 03/2013-03/2015 | Department of Engineering Mechanics, SJTU
Research Assistant in <i>Computational Fluid Dynamics</i> <ul style="list-style-type: none">● Lead a project on flow characteristics around spur dikes in an open channel● Rich experience in computer programming, algorithms, and numerical analysis |

- Enabled large scale simulations on multiple CPUs via OpenMP
- 06/2013-03/2014 Ministry of Education (MOE) Key Laboratory of Hydrodynamics, SJTU
Research Assistant in *Particle Image Velocimetry (PIV) Experiment*
 - Independently conducted a PIV experiment to measure the flow field
 - Assisted in a project on flow measurements around bridge piers
- 09/2011-06/2012 Department of Engineering Mechanics, WUST
Research Assistant in *Engineering Mechanics*
 - Assisted in a project on carbon fiber reinforced concrete design
 - Gained the solid knowledge of Finite Element Method (FEM)

PUBLICATIONS

1. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. Under review in the *AIAA Journal*. Doi: 10.48550/arXiv.2511.18276
2. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Accurate Predictions of Time Resolved Unsteady Flows Over Airfoils. *AIAA SciTech Forum*, 2025. Doi: 10.2514/6.2025-2049
3. **Z. Li**, P. He. Accelerating Unsteady Aerodynamic Simulations Using Predictive Reduced order Modeling. *Aerospace Science and Technology*, 2023. Doi: 10.1016/j.ast.2023.108412
4. **Z. Li**, P. He. Airfoil Unsteady Aerodynamic Analysis Using a Galerkin Reduced order Modeling Approach. *AIAA SciTech Forum*, 2022. Doi: 10.2514/6.2022-0080
5. **Z. Li**, J. Kou, & J. Zhang. Numerical simulations of flows around a single spur dike in an open channel. *Chinese Journal of Computational Mechanics*, 2016. Doi: 10.7511/jslx201602016

TEACHING EXPERIENCE

- 01/2022-Present Department of Aerospace Engineering, ISU
Teaching Assistant in *Aerospace Systems Integration, Advanced Flight Structures, Design of Aerospace Systems, Modern Design Methodology with Aerospace Applications*

COMPUTER SKILLS

Programming Languages: Strong proficiency in C/C++, Python, Fortran, MATLAB

CFD Packages: Extensive experience with OpenFOAM, Fluent

Operating Systems: Expertise in Linux (Ubuntu, Fedora)

AWARDS & HONORS

- 2025 Research Excellence Award in the College of Engineering, ISU
- 2018 H. Marcus Dean's Chair Scholarship in the College of Engineering, PSU
- 2018 Harry G. Miller Scholarship in the College of Engineering, PSU
- 2017 University Graduate Fellowship (UGF) in the College of Engineering, PSU
- 2014 2nd Prize Scholarship (Top 15%), SJTU
- 2013 1st Prize in National Graduate Mathematical Modeling Contest (Top 3%)
- 2012 Outstanding Graduate Student Award (Top 10%), WUST
- 2011 3rd Prize in National Undergraduate Mathematical Contest in Modeling (Twice)
- 2010 National Encouragement Scholarship (Top 5%), WUST
- 2010 3rd Prize in National College Student Basic Mechanics Experiment Contest (Top 3%)

2010 “Excellent” Prize in National Pei-Yuan Chou Mechanics Competition

EXTRACURRICULAR ACTIVITIES

2013-2014 **Assistant**, Employment Guidance Center, SJTU

2010-2012 **Vice-Chairman**, Mathematical Modeling Association, WUST

2009-2012 **Core Member**, Wuhan University of Science and Technology Table Tennis Club